

- (i) Lei 'n uitdrukking af vir die snelheid v en versnelling a in terme van 'n Sin- of Cos-funksie.
Find an expression for the velocity v and the acceleration a in terms of a Sin- or Cos-function.

$$v = -A\omega \sin \omega t$$

$$a = -A\omega^2 \cos \omega t$$

(2)

- (ii) Skryf nou die versnelling in terme van x . / Write the acceleration in terms of x .

$$a = -\omega^2 x$$

(1)

- (c) Toon aan dat vir die massa m aan die veer geld $\omega = \sqrt{\frac{k}{m}}$.

Show that for mass m on a spring $\omega = \sqrt{\frac{k}{m}}$.

$$F = ma = -kx$$

$$a = \frac{-kx}{m}$$

$$= -\omega^2 x$$

$$\therefore \omega = \sqrt{\frac{k}{m}}$$

(4)

- (d) Gee 'n uitdrukking vir die periode T in terme van m en k .
Give an expression for the period T in terms of m and k .

$$T = 2\pi \sqrt{\frac{m}{k}}$$

(2)

- (e) Wat is die verband tussen periode T en frekwensie f ?
What is the relation between period T and frequency f ?

$$T = \frac{1}{f} \quad f = \frac{1}{T}$$

(1)

- (f) Met $x = A \cos \omega t$, toon aan dat die potensiële energie van 'n massa m aan 'n veer gegee word deur

With $x = A \cos \omega t$, show that the potential energy of a mass m on a spring is given by

$$U = \frac{1}{2} k A^2 \cos^2 \theta.$$

$$U = \frac{1}{2} k x^2$$

$$= \frac{1}{2} k (A \cos \omega t)^2$$

$$= \frac{1}{2} k A^2 \cos^2 \omega t$$

(2)